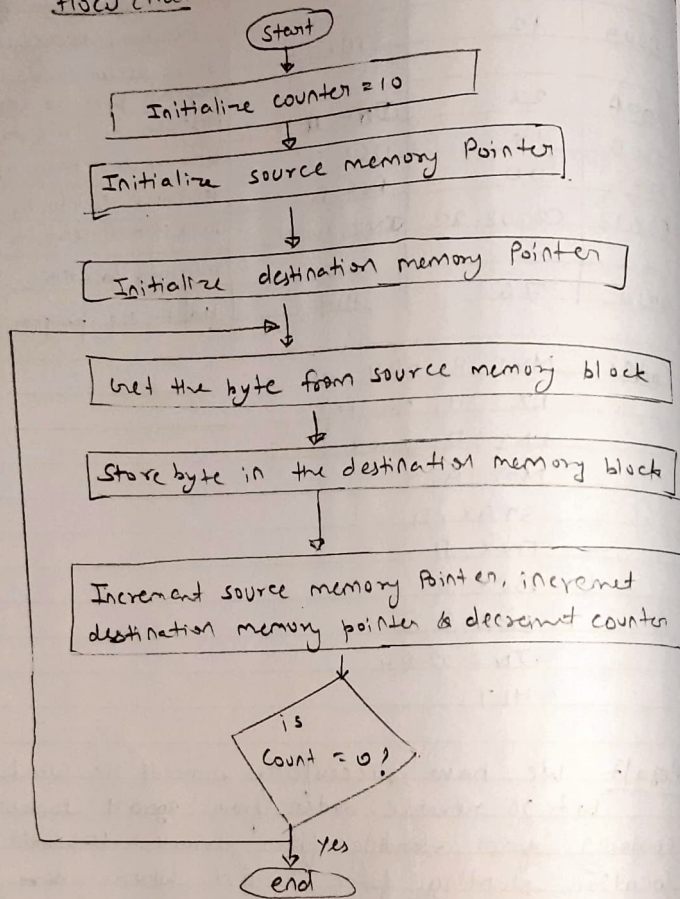


flow chart



Aim :- Program to write swap two blocks of data stored in memory.

Apparatus Required - +5volt SMPS, Microprocessor 8085 Dyna-kit

Program Analysis -

1. Start the program
2. Initialize register pair DE
→ Loading starting address of first block
3. Initialize register pair HL
→ Loading starting address of second block
4. Initialize Counter (C) register
→ set count = 4 (number of data elements)
5. Repeat the following steps until count = 0;
 - (a) Move content of memory (HL) into register B
 - (b) Load content of memory (DE) into accumulator
 - (c) Store accumulator content into memory (HL) (A)
 - (d) Move content of register B into Accumulator
 - (e) Store accumulator content into memory (DE)
 - (f) Increment HL & DE
 - (g) Decrement counter C
6. Check if C $\neq$ 0, repeat steps 5
7. Stop program.

OutputInput

C050 = 0A

C051 = 13

C052 = 08

C053 = 0C

~~C054~~

C060 = 12

C061 = 11

C062 = 10

C063 = 09

/

Output

C050 = 12

C051 = 11

C052 = 10

C053 = 09

C060 = 0A

C061 = 13

C062 = 0B

C063 = 0C

Program -

LXI D, C050

LXI H, C060

MVI C, 04

MOV B, M 46

LDAX D

MOV M, A

MOV A, B

STAX D

INX H

INX D

DCR C

JNZ C008

HLT

Result we have successfully performed the swapping of two blocks.

Aim - Write a program to find the square of a number.

Apparatus Required:- +5 Volt SMPS, Microprocessor 8085 Dyna-kit.

Program Analysis -

1. Start Program

2. Initialize register D = 00H

→ This will store carry (higher byte of result)

3. Load HL with address C500

→ Point to input data.

4. Load data from memory into registers.

→ Move memory content into register C (Counter)

→ Move some data into accumulator A

5. Decrement counter ($C = C - 1$)

6. Repeat the following steps -

(a) Add memory content (M) to accumulator A

(b) If carry occurs → increment register D.

(c) Decrement counter C

(d) If $C \neq 0$ → repeat loop

7. Store result

→ Store Accumulator (A) → C601

Move carry (D) → A

Store A → C602

8. Stop the program.

Output

input

C500 = 02

output Value

C601 = 00

C602 = 04

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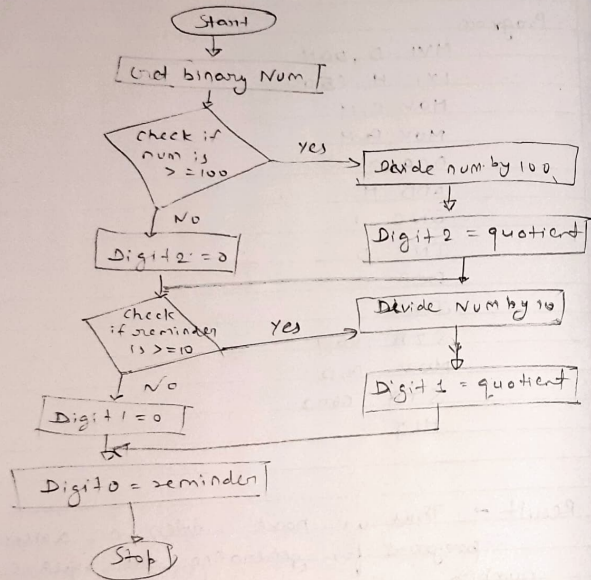
Date _____

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Program

```
MVI D, 00H
LXI H, C500
MOV C, M
MOV A, M
DCR C
ADD M
JNC I
INR D
DCR C
JNZ
STA C601
MOV A, D
STA C602
HLT
```

Result - Thus we have written an assembly language program for generating the square of a given number.



Aim:- Write a program for Binary to BCD conversion

Apparatus Required:- +5 Volt, SMPs, Microprocessor 8085 Dyna-kit

Program Analysis:-

1. Start program
2. Load input data
 - Load HL with address C050
 - Move memory content into accumulator (A)
3. Call subordinate Subroutine PWRTRN
4. Inside PWRTRN:
 - Initialize HL to C070 (output location)
 - Load B = 64H (100 in decimal)
5. Find Hundreds digit
 - Call subroutine BINBCD
 - Store quotient (Hundreds digit)
6. Find tens digit
 - Load B = 0AH (10 decimal)
 - Call BINBCD again
7. Store units digit
 - Remaining value in A is units digit
 - Store it in memory
8. Stop program.

input

C050 = 21

output

C070 = 00

C071 = 03

C072 = 03

Program

LXI H, C050

MOV A, M

CALL PWRTEN

HLT

LXI H, C070

MVI B, 64H

CALL BINBCD

MVI B, 0AH

CALL BINBCD

MOV M, A

RET

MVI M, FEH

INR M

SUB B

JNC NXT

ADD B

INX H

RET

Result The program to convert the binary number to BCD number has been done successfully.

Object 1:- Write a program to find largest & smallest number from a given array.

Apparatus Required :- 45 Volt SMPS, Microprocessor 8085 Dyna Kit.

Steps for finding the large & smaller number

1. Initialize memory pointer & initialize counter C
2. Read number for the list store in B
3. If A is greater than B
4. Store large number in A
5. Increment the counter
6. Decrement counter C
7. If A is greater than B
8. Display the highest number
9. Read number from the list

Program - for finding ~~smallest~~ number largest number

LXI H F500

MOV C, M

INX H

MOV A, M

DCR C

INX H

MOV B, M

CMP B

JNC F00E

MOV A, B

DCR C

JNZ F00F

STA F601

HLT

Teacher's Signature _____

Input

F500 = 04

F501 = 07

F502 = 03

F503 = 0A

Output

F601 = 0A

input Values

F500 = 04

F501 = 07

F502 = 03

F503 = 0A

output

F601 = 03.

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for finding largest number, smallest

LXI H, F500

MOV C, M

INX H

MOV A, M

INX H

~~MOV A,~~

DCR C

INX H

MOV B, M

CMP B

JC F00E

MOV A, B

DCR C

JNZ F007

STA F601

HLT

Teacher's Signature _____